

Important Topics to Cover in Low Level Design (LLD)

Core OOP Concepts

- Classes & Objects
- Encapsulation
- Abstraction
- Inheritance
- Polymorphism
- Composition vs Inheritance
- SOLID Principles

UML & Design Basics

- Class Diagrams
- Sequence Diagrams
- Use Case Diagrams
- Association, Aggregation, Composition

Design Patterns

- Singleton
- Factory
- Abstract Factory
- Builder
- Strategy
- Observer
- Decorator
- Adapter
- State
- Command

Clean Code & Best Practices

- DRY Principle
- KISS Principle
- YAGNI
- Loose Coupling

- High Cohesion
- Modular Design

Concurrency & Multithreading

- Threads
- Synchronization
- Locks
- Deadlocks
- Thread Safety
- Executor Framework

API & Service Design

- REST APIs
- DTOs
- Validation
- Error Handling
- Pagination
- Rate Limiting

Database-Oriented LLD

- Entity Relationships
- ORM Concepts
- Transactions
- Indexing
- Repository Pattern

Machine Coding Problems

- Parking Lot
- Elevator System
- Splitwise
- BookMyShow
- Cab Booking
- Food Delivery
- ATM System

Advanced LLD Topics

- Dependency Injection
- Caching

- Pub/Sub
- Retry Mechanisms
- Circuit Breaker
- Rate Limiter Design

Java-Specific LLD Topics

- Interfaces vs Abstract Classes
- Comparable vs Comparator
- equals() & hashCode()
- Immutable Classes
- Java Collections Framework
- Streams API

Most Important Topics for Interviews

- SOLID Principles
- Design Patterns
- OOP Modeling
- Machine Coding Problems
- Clean Code
- UML Diagrams
- Concurrency Basics